Geon-Woo Kim

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Summary

Researcher and engineer building **large-scale ML systems and infrastructure** for LLM training and serving, accelerator collective communication, fault tolerance, and performance. Ph.D. candidate at UT Austin (graduating Nov 2026) with publications at OSDI, SOSP, ICML, NeurIPS, and IPDPS, and research and engineering experience at Google and Meta. Previously spent five years in industry building and operating production platforms serving 20M+ users.

Education

The University of Texas at Austin — Austin, TX, USA
Ph.D. in Computer Science — *Nov 2026 (Expected)*
M.S. in Computer Science — *Dec 2025*
Advisor: *Aditya Akella and Daehyeok Kim*

Seoul National University — Seoul, South Korea
B.S. in Computer Science & Engineering; B.S. in Mathematical Sciences (Double Major) — *Mar 2013 - Aug 2022*
- Summa Cum Laude. Completed both degrees concurrently with five years of full-time industry experience and alternative military service.

Experience

Graduate Research Assistant, UTNS — UT Austin
Advisor: *Aditya Akella and Daehyeok Kim* — *Sep 2022 - Nov 2026 (Expected)*
- *Belay (in submission)*: Building a fault-tolerance system for large-scale LLM training that recovers from *transient* GPU, NIC, and network faults without full-job restarts, at negligible steady-state overhead.
- *HALoS (ICML 25)*: Built a hierarchical asynchronous optimization framework for geo-distributed LLM training in collaboration with Meta; converges **up to 7.5× faster** than synchronous training and **2.1× faster** than prior asynchronous methods while matching model quality, with formal convergence guarantees.
- *Reforge (IPDPS 26)*: Led the design of a distributed GNN serving system that cuts serving latency by **up to 159×** over full-graph serving and **10.8×** over approximation baselines with <1% accuracy loss, via selective embedding recomputation and computation-graph parallelism.

Software Engineer Intern, Machine Learning (PhD) — Meta Platforms, Inc.
Mentor: *Saif Hasan* (AI Networking Team) — *May 2025 - Aug 2025*
- Developed **NVIDIA–AMD heterogeneous GPU collectives** (AllReduce, AllToAll) in NCCLX, Meta's collective communication library (open-sourced as torchcomms), across RDMA NICs and NVLink/xGMI interconnects.
- Made **100%** of the NCCLX codebase compilable on both CUDA and ROCm with single-source builds.
- Built an RDMA performance tool for mixed-vendor environments (NVIDIA–AMD GPUs, Mellanox–Broadcom NICs).

Student Researcher, SystemsResearch@Google — Google
Mentor: *Seo Jin Park* — *May 2023 - Jul 2023*
- *FluidEP (SOSP 26, to appear)*: Proposed and simulated fine-grained expert disaggregation for MoE LLM serving; the project continued with collaborators and culminated in a barrier-free expert-parallel decoding system that achieves **up to 1.7× higher throughput** than SGLang on commodity GPU clusters.

Undergraduate Research Intern, SPL — SNU
Advisor: *Byung-Gon Chun* — *Jan 2015 - Aug 2021*
- *Orca (OSDI 22; 1,300+ citations)*: Core contributor to the serving system that introduced ***continuous batching***, now the standard technique adopted in vLLM, TensorRT-LLM, and SGLang; extended attention CUDA kernels to enable fine-grained KV cache management; and built the end-to-end serving path, from the low-latency C++ web frontend to distributed multi-GPU deployment.
- Integrated NVIDIA CUPTI into Orca's C++ engine and enabled browser-based GPU kernel-trace visualization for performance profiling and debugging.
- *Terra (NeurIPS 21)*: Enabled symbolic execution of imperative TensorFlow programs; strengthened formal guarantees with JIT compilation and mathematical proof, and extensively profiled and optimized the system.

Software Platform Engineer — Viva Republica (Toss)
Korea's leading fintech, valued at $7B+; joined as the 3rd server developer (100+ today) — *Jun 2016 - Feb 2021*

- Proposed and drove **TUBA**, Toss's company-wide mobile marketing platform for A/B testing, large-scale targeted messaging, and real-time, log-driven in-app prompts, serving **20M+ users**; collaborated with data engineers and app/server developers to deliver it end-to-end.
- Operated TUBA in production for **3+ years**; the platform is now owned by its own dedicated team.
- Built and maintained a large-scale data analysis platform on Apache Spark, HDFS, Impala, Flink, and PyTorch.

Technical Skills

Languages: Python, C++, C, Java, Kotlin
GPU Computing & Communication: CUDA, ROCm, NCCL/RCCL, CUPTI, RDMA (InfiniBand/RoCE)
ML Systems: PyTorch, TensorFlow, vLLM, DeepSpeed, TorchTitan, Megatron-LM, DGL

Publications

[1] **FluidEP: Democratizing MoE LLM Decoding via Barrier-Free Expert Parallelism**, Yizhuo Liang, Shaoyu Wang, Jaeyong Song, Yanqi Zhou, *Geon-Woo Kim*, Guangrong He, Seo Jin Park, ACM Symposium on Operating Systems Principles (**SOSP 26**, to appear)

[2] **Reforge: Low-Latency Distributed GNN Serving with Selective Embedding Recomputation**, *Geon-Woo Kim*, Donghyun Kim, Jeongyoon Moon, Henry Liu, Tarannum Khan, Anand Iyer, Daehyeok Kim, Aditya Akella, IEEE International Parallel and Distributed Processing Symposium (**IPDPS 26**)

[3] **Dooly: Configuration-Agnostic, Redundancy-Aware Profiling for LLM Inference Simulation**, Joon Ha Kim, *Geon-Woo Kim*, Anoop Rachakonda, Daehyeok Kim, arXiv preprint arXiv:2605.07985 (**Preprint**)

[4] **HALoS: Hierarchical Asynchronous Local SGD over Slow Networks for Geo-Distributed Large Language Model Training**, *Geon-Woo Kim*, Junbo Li, Shashidhar Gandham, Omar Baldonado, Adithya Gangidi, Pavan Balaji, Zhangyang Wang, Aditya Akella, Forty-second International Conference on Machine Learning (**ICML 25**)

[5] **StitchLLM: Serving LLMs, One Block at a Time**, Bodun Hu, Shuozhe Li, Saurabh Agarwal, Myungjin Lee, Akshay Jajoo, Jiamin Li, Le Xu, *Geon-Woo Kim*, Donghyun Kim, Hong Xu, Amy Zhang, Aditya Akella, The 63rd Annual Meeting of the Association for Computational Linguistics (**ACL 25**)

[6] **Read-ME: Refactorizing LLMs as Router-Decoupled Mixture of Experts with System Co-Design**, Ruisi Cai, Yeonju Ro, *Geon-Woo Kim*, Peihao Wang, Babak Ehteshami Bejnordi, Aditya Akella, and Zhangyang Wang, Thirty-Eighth Annual Conference on Neural Information Processing Systems (**NeurIPS 24**)

[7] **Lovelock: Towards Smart NIC-hosted Clusters**, Seo Jin Park, Ramesh Govindan, Kai Shen, David Culler, Fatma Özcan, *Geon-Woo Kim*, and Hank Levy, HotCarbon Workshop on Sustainable Computer Systems (**HotCarbon 24**)

[8] **Orca: A distributed serving system for Transformer-Based generative models**, Gyeong-In Yu, Joo Seong Jeong, *Geon-Woo Kim*, Soojeong Kim, and Byung-Gon Chun, USENIX Symposium on Operating Systems Design and Implementation (**OSDI 22**)

[9] **Terra: Imperative-Symbolic Co-Execution of Imperative Deep Learning Programs**, Taebum Kim, Eunji Jeong, *Geon-Woo Kim*, Yunmo Koo, Sehoon Kim, Gyeong-In Yu, and Byung-Gon Chun, Thirty-Fifth Annual Conference on Neural Information Processing Systems (**NeurIPS 21**)

[10] **Apache Nemo: A Framework for Building Distributed Dataflow Optimization Policies**, Youngseok Yang, Jeongyoon Eo, *Geon-Woo Kim*, Joo Yeon Kim, Sanha Lee, Jangho Seo, Won Wook Song, and Byung-Gon Chun, USENIX Annual Technical Conference (**ATC 19**)

[11] **Pado: A data processing engine for harnessing transient resources in datacenters**, Youngseok Yang, *Geon-Woo Kim*, Won Wook Song, Yunseong Lee, Andrew Chung, Zhengping Qian, Brian Cho, and Byung-Gon Chun, ACM European Conference on Computer Systems (**EuroSys 17**)

Professional Service

2026 **Reviewer (NeurIPS)** Advances in Neural Information Processing Systems
2026 **Reviewer (TPDS)** IEEE Transactions on Parallel and Distributed Systems
2026 **Artifact Evaluation Committee (OSDI)** Symposium on Operating Systems Design and Implementation

Honors & Awards

Aug 2022 **Overseas PhD Scholarship** — Korea Foundation for Advanced Studies
Mar 2013 **Presidential Science Scholarship** — Korean Government